Notes_25.14: Cross Validation for Hyperparameter Tuning (GridSearchCV)







🌐 **Language: English** | हिंदी / Hinglish | ⬅️ Lecture 25 Overview

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📌 Executive Summary

Exhaustive grid search with cross-validation to select optimal K and distance metrics.

🎯 Mathematical Formulations

Optimal Hyperparameter Selection:

$$ \theta^* = \arg\max_{\theta \in \Theta} \frac{1}{K} \sum_{k=1}^K \text{Metric}\left(f_{\theta}(\mathcal{D}{\text{train}}^{(-k)}), \mathcal{D}{\text{val}}^{(k)} \right) $$

💡 Conceptual Principles & Architecture

GridSearchCV Kaise Kaam Karta Hai?

Hum hyperparameters ki ek dictionary pass karte hain (jaise `n_neighbors`: $[3, 5, 7, 9, 11]$ aur `weights`: `['uniform', 'distance']`). GridSearchCV har combination par K-Fold Cross Validation run karta hai aur mathematically sabse best hyperparameter θ^* select kar leta hai!

💻 Python Production Implementation

```
from sklearn.model_selection import GridSearchCV
from sklearn.neighbors import KNeighborsClassifier
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
import pandas as pd

df = pd.read_csv("heart.csv")
X = df.drop("target", axis=1)
y = df["target"]

pipe = Pipeline([
    ('scaler', StandardScaler()),
    ('knn', KNeighborsClassifier())
])

param_grid = {
    'knn__n_neighbors': [3, 5, 7, 9, 11, 15],
    'knn__weights': ['uniform', 'distance'],
    'knn__p': [1, 2]  # 1=Manhattan, 2=Euclidean
}

grid = GridSearchCV(pipe, param_grid, cv=5, scoring='accuracy')
grid.fit(X, y)

print("Best Parameters:", grid.best_params_)
print(f"Best CV Accuracy: {grid.best_score_:.4f}")
```